Sofia Duarte

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Education

2023 – Exp. 2026	**BSc in Aerospace Engineering** at **Instituto Superior Técnico - University of Lisbon** (Ongoing)
2020 – 2023	**High School Diploma** (Grade: 19.11/20)

Work Experience

RED - Electronics Co-Team Leader June 2025 - present

- **Flight Software Architecture:** Architected the division's first Hybrid Rocket flight software running on an RTOS. Implemented critical control logic for propulsion valves, ignition systems, and recovery actuators.
- Leading software design for RED's first distributed avionics system and an automatic filling station; Developing Hardware-in-the-Loop (HIL) testing software in collaboration with the Control Team to validate performance and aiding in integrating AHRS filters (IMU/GNSS fusion).
- **Rust Pilot Software Project (Inclita Geração):** Pioneering the adoption of Rust for flight software within the RED. Architecting the flight software for the "Inclita Geração" CANSATs launcher rockets in collaboration with the R&D team.
- **Technical Leadership:** Supervise the electronics team, conducting code reviews and mentoring members. Act as a bridge between hardware and software teams for cohesive system design.
- **Systems Engineering:** Manage interface control between Electronics and other departments; developing the team's first software Documentation Database and establishing CI/CD workflows via GitHub Actions.
- **Test Infrastructure:** Refactored legacy software for the Solid Motor Test-bench to optimize SD-card logging latency; developed software for Hybrid Static Fire tests for data acquisition and execute remote actuation.

RED - Electronics Team Member October 2024 - May 2025

- **Embedded Drivers:** Developed bare-metal firmware drivers using I2C, UART, SPI, and SDIO protocols to interface STM32 microcontrollers with sensors and other peripherals.
- **Debugging Tools:** Investigated new firmware solutions for embedded ST-LINK devices to optimize the programming and debugging workflow for the team's hardware.

RED - Safety Team Member June 2025 - present

- Developed Checklist and evaluated safety hazards regarding software and hardware to mitigate risks during static fire tests and launch operations.
- Completed certified training in Basic Life Support (BLS) and fire extinguishing techniques provided by firefighting services.

LISAT - Command and Data Handling Department Member March 2025 - August 2025

- **Memory Optimization:** Developed a cache optimization algorithm to streamline the transfer of orbit data from external RAM to main memory, improving data processing throughput for the satellite's onboard computer

Projects

Rust Pilot Software Project (WIP) Flight Software Repo

Architecting deterministic, asynchronous flight software for an STM32 microcontroller entirely in embedded Rust, implementing hardware-level interrupt pre-emption to guarantee deterministic sensor control loops and data recording.

Rust-based HIL & Ground Station Framework (WIP) HIL Repo | GS Dashboard Repo
Architecting a modular, high-performance simulation and telemetry engine to validate avionics control logic, to integrate with MATLAB. Developed a unified backend in Rust (Async/Await) designed for memory safety and low latency. The framework aims to fuse Hardware-in-the-Loop (HIL) framework to interface with the Ground Station dashboard, utilizing a protocol-agnostic communication layer to toggle between simulated environments via Wiring and physical hardware via during flight. Focused on the transition from a C codebase to a Rust codebase.

Technical Skills

Systems & Embedded	**Languages:** C, C++, Rust, RISC-V Assembly **Architecture:** Dual-Core Processing, Bare-Metal, Concurrent Programming **Frameworks & RTOS:** Embassy (Async Rust), FreeRTOS **Platforms:** STM32, ESP32, Arduino **Electronics:** KiCad (Basic PCB Design), Soldering
Scientific & Analysis	**Languages:** Python, MATLAB, Wolfram Language **Focus:** Data Analysis, Simulation, Modeling
Tools & Automation	**Linux:** Scripting (Bash, Lua), Environment Config **CI/CD & Docs:** Git (GitHub Actions – YAML Workflows, GitHub Pages), Doxygen, Latex **Web Fundamentals:** HTML, CSS, JavaScript

Language Skills

Portuguese:	Native Language
English:	Advanced Speaker (IELTS Academic Overall Band 8.0)

Soft Skills

Leadership & Teamwork	**Electronics Team Lead:** Experience managing hardware design and integration. Participated in the EuRoc 2025 Competition.
Problem Solving	**Root Cause Analysis:** Debugging complex mixed-signal systems. Troubleshooting concurrency issues (Deadlocks, Race Conditions) in real-time systems.
Communication	**Intercultural Adaptability:** Strong communication skills fostered by international living experience. Effective in diverse, multi-lingual environments.